

Employee Attrition Prediction – Summary

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Project: Employee Attrition Prediction using Machine Learning

Project Overview

The objective of this project was to identify employees who are most likely to leave the company and understand the factors contributing to employee attrition. By analyzing historical employee data, the project aimed to provide practical insights that can help the HR team improve employee retention and reduce recruitment and training costs.

Key Findings

The analysis showed that employee attrition is influenced by several workplace factors rather than salary alone.

- The **Sales department** had the highest attrition rate (**20.6%**), while **Research & Development** had the lowest (**13.8%**).
- Among job roles, **Sales Representatives (39.8%)**, **Laboratory Technicians (23.9%)**, and **Human Resources employees (23.1%)** experienced the highest turnover.
- Employees with the lowest **Work-Life Balance** rating had the highest attrition rate (**31.2%**), indicating that work-life balance plays an important role in employee retention.
- The highest employee turnover occurred during the **first two years** of employment (**29.8%**). Employees with **11–20 years** of service had the lowest attrition rate (**6.7%**).
- Although employees who left generally earned lower salaries, overtime, business travel, delayed promotions, and specific job roles had a greater influence on employee attrition.

Recommendations

HR should prioritize retention efforts in the **Sales department**, especially for **Sales Representatives** and **Laboratory Technicians**. Employees who frequently work overtime or travel for business should receive regular support, workload reviews, and work-life balance initiatives to reduce burnout.

The company should also strengthen career development by providing timely promotions, regular career discussions, and mentoring programs. Special attention should be given to employees during their **first two years**, as this is the period with the highest risk of attrition.

Conclusion

This project demonstrates that employee data can help identify individuals who may be at risk of leaving the company before they resign. These insights can support HR in taking proactive steps to improve employee satisfaction and retention. However, the model should be used as a **decision-support tool** rather than the only basis for HR decisions. Combining data-driven insights with employee feedback and managerial judgment will help build a more engaged, productive, and stable workforce.